Exam 2 Review – PSTAT 5LS

- Slide Sets 5a, 5b, 6, 7, 8
- HW 3-5

Learning Objectives

To do well on Exam 2, you should be able to:

- Describe the distribution (shape, center, variability/spread, and outliers) of a variable from a histogram
- Identify is a study is an observational study or an experiment
- Determine which is the explanatory and which is the response variable
- Identify the parameter of interest (p, μ, $\mu_1 - \mu_2$, μ_d) in a scenario
- Define a parameter in context
- Determine if we need to use the *z* or t for inference
- State the null and alternative hypotheses
- Hypothesis tests for p, μ, $\mu_1 - \mu_2$, μ_d:
    - State the null and alternative hypotheses
    - Define the parameter in context
    - Check the conditions
    - Calculate the test statistic
    - Determine which R function will be used to calculate a p-value
    - Explain how the magnitude of the test statistic is related to the p-value
    - Determine whether you should reject or fail to reject the null hypothesis
    - Determine if the results are statistically significant
    - Make a conclusion in context
- Confidence intervals for p, μ, $\mu_1 - \mu_2$, μ_d:
    - Define the parameter in context
    - Check the conditions
    - Determine which R function will be used to find a critical value
    - Calculate a confidence interval
    - Interpret a confidence interval in context

Use the following information for Questions 1-4:
Can self-help mobile messaging interventions improve depressive symptomatology (the grouping of symptoms that categorize depression) in older adults (60+ years)? Researchers in Brazil recently published a study that suggests this is the case. One of the concerns of the researchers was that older adults who live in areas with few health professionals who cannot take on additional course loads might not have access to care to help their depressive symptomatology. Here's where low-cost self-help digital mental health interventions can help.

A total of 603 participants were involved in the study from beginning to end. These participants were randomly assigned into two groups, with 298 in the intervention group and 305 in the control group.

All participants took a 9-item Patient Health Questionnaire (PHQ-9) at the beginning of the study. Those participants in the intervention group used the self-help mobile messaging system called Viva Vida ("Live Life" in English) for 6 weeks, with no additional health professional support. Those in the control group were "were sent a single and brief (6-min) audio message providing information about depression and simple ways to deal with depressive symptoms. It also offered additional tips encouraging participants to live a healthier life, such as improving sleep and maintaining a balanced diet, advising them to seek health care if they felt they needed additional support."[1]

After 3 months, all participants still involved in the study (257 in the intervention group, 270 in the control group) took the PHQ-9 again. At this time, 42.4% of those in the intervention group showed improvement in their PHQ-9 score, compared to 32.2% in the control group.

1. This research was an
   a. experiment.
   b. observational study.

2. The variable "difference in PHQ-9 scores (before – after)" is a
   a. quantitative variable.
   b. categorical variable.

3. The variable "type of intervention" is the
   a. explanatory variable.
   b. response variable.

4. The randomization of the participants into the two study groups means that
   a. the difference in PHQ-9 scores (before – after) are independent for all of the participants.
   b. the results from this research can be generalized to the population of older adults in Brazil.
   c. the treatment groups were as similar as possible other than the treatment.

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[1] Scazufca, M., Nakamura, C.A., Seward, N. *et al.* Self-help mobile messaging intervention for depression among older adults in resource-limited settings: a randomized controlled trial. *Nat Med* (2024). https://doi.org/10.1038/s41591-024-02864-4

Use the following information for Questions 5-8:
Different species can interact in interesting ways. One type of grass produces the toxin ergovaline at levels about 1.0 part per million in order to keep grazing animals away. However, a study[2] found that the saliva from a moose counteracts the toxins in this type of grass and makes the grass more appetizing (for the moose). Scientists reported a 95% confidence interval of 0.167 to 0.199 for the mean level of the toxin ergovaline (in ppm) on the grass, after treatment with moose drool. Identify whether each of the statements below is a correct statement regarding this interval.

5. A 90% confidence interval calculated from the same data would be narrower than the given interval. This statement is
   a. correct.
   b. incorrect.

6. We can be 95% confident that the population mean level of ergovaline (in ppm) is between 0.167 and 0.199. This statement is
   a. correct.
   b. incorrect.

7. There is a 95% chance that the population mean level of ergovaline is between 0.167 ppm and 0.199 ppm. This statement is
   a. correct.
   b. incorrect.

8. Approximately 95% of all measurements of ergovaline (in ppm) for this type of grass will fall between 0.167 ppm and 0.199 ppm. This statement is
   a. correct.
   b. incorrect.

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9. A random sample of students in a student success program was taken, and a 95% confidence interval for the mean GPA (grade point average) for all students in program was calculated. The resulting confidence interval was 3.44 to 3.50. Which of the following is **not** true?
   a. The sample mean GPA was 3.47.
   b. The probability that the population mean GPA for students in the student success program is in the confidence interval is 0 or 1, but we don't know which.
   c. The confidence interval provides a range of plausible values for the population mean GPA for students in the student success program.
   d. The confidence interval provides a range of plausible values for the sample mean GPA for students in the student success program.

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10. What is the difference between t^* and the test statistic t?
    a. t^* assumes the null hypothesis is true, and the test statistic t assumes the alternative hypothesis is true.
    b. t^* is determined by the confidence level, and the test statistic t is calculated from the data.
    c. The test statistic t assumes the null hypothesis is true, and t^* assumes the alternative hypothesis is true.
    d. The test statistic t is determined by the confidence level, and t^* is calculated from the data.

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[2] Tanentzap, A.J., Vicari, M., Bazely, D.R., "Ungulate saliva inhibits a grass-endophyte mutualism," Biology Letters, 10(7), July 2014.

Use the following information to answer Questions 11-14:
For each of the following situations, identify the parameter of interest.

11. A university officer wants to estimate the average amount of time that students spend in the campus library. The parameter is
    a. p.
    b. μ.
    c. $\mu_1 - \mu_2$.
    d. μ_d.

12. Researchers studying cluster headaches recruit 25 patients who suffer from cluster headaches and 25 patients who do not suffer from cluster headaches. The researchers want to know if there is a difference in the average vitamin D level for people who suffer from cluster headaches and people who do not suffer from cluster headaches. They should
    a. p.
    b. μ.
    c. $\mu_1 - \mu_2$.
    d. μ_d.

13. Data scientists at an online retailer want to estimate the percentage of people who visit the site who make a purchase. The parameter is
    a. p.
    b. μ.
    c. $\mu_1 - \mu_2$.
    d. μ_d.

14. Researchers want to determine if practicing mindfulness is helpful for lowering resting pulse rates. The recruit 20 participants for their study and measure each participant's resting pulse rate. The participants are then instructed about some basic mindfulness exercises and are told to spend 15 minutes a day using at least one of the mindfulness exercises. One week later, the participants come back in, and the researchers measure each participant's resting pulse rate. The parameter is
    a. p.
    b. μ.
    c. $\mu_1 - \mu_2$.
    d. μ_d.

Use the following information to answer Questions 15-19:

An Associated Press-NORC Center for Public Affairs Research publication[3] was titled "Most say democracy is important for the U.S. identity, but few think it is functioning well." In a March 2024 poll with a random sample of size 1282 adults living in the United States, 679 responded that they think that democracy is not functioning well in the United States. Let *p* represent the proportion of all U.S. adults who think that democracy is not functioning well in the United States.

15. Which of the following is the point estimate for the proportion of U.S. adults who think that democracy is not functioning well in the United States?
    a. $p = 0.530$
    b. $\hat{p} = 0.530$
    c. $x = 679$
    d. $\bar{x} = 679$

16. Which of the following conditions need to be checked before we construct a confidence interval? (Select all that apply.)
    a. The observations are independent of one another.
    b. $\boldsymbol{n \geq 30}$
    c. $\boldsymbol{n\hat{p} \geq 10}$ and $\boldsymbol{n(1-\hat{p}) \geq 10}$
    d. $\boldsymbol{n\hat{p} \geq 30}$ and $\boldsymbol{n(1-\hat{p}) \geq 30}$

17. Which of the following do we need to use to find the critical value for a 98% confidence interval?
    a. `qnorm(0.01, mean = 0, sd = 1, lower.tail = FALSE)`
    b. `qnorm(0.02, mean = 0, sd = 1, lower.tail = FALSE)`
    c. `qt(0.01, df = 79, lower.tail = FALSE)`
    d. `qt(0.02, df = 79, lower.tail = FALSE)`

18. The 98% confidence interval for the proportion of all U.S. adults who think that democracy is not functioning well in the United States is (0.497, 0.562). Which of the following must be the 95% confidence interval?
    a. 0.494 to 0.566
    b. 0.497 to 0.552
    c. 0.502 to 0.557
    d. 0.507 to 0.562

19. As mentioned above, the 98% confidence interval for the proportion of all U.S. adults who think that democracy is not functioning well in the United States is (0.497, 0.562). Does this confidence interval suggest that a majority (more than 50%) of U.S. adults think that democracy is not functioning well in the United States?
    a. Yes, because most of the values in the confidence interval are more than 0.50.
    b. No, because most of the values in the confidence interval are more than 0.50.
    c. Yes, because some of the values in the confidence interval are less than 0.50.
    d. No, because some of the values in the confidence interval are less than 0.50.

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[3] https://apnorc.org/projects/most-say-democracy-is-important-for-the-u-s-identity-but-few-think-it-is-functioning-well/

20. **Cholesterol Levels** The CDC reported that the average total cholesterol level in U.S. adults was 189 mg/dL in 2013-2014. Data have shown a decreasing trend in total cholesterol level in the population of U.S. adults since the 1980s. Recently, a researcher wanted to investigate whether the average total cholesterol level for U.S. adults has decreased from the 2013-2014 level. The researcher recruited a random sample of 80 individuals and found their mean total cholesterol level to be 184.9 mg/dL with a standard deviation of 46.2 mg/dL.



a. State the hypotheses that the researcher should use to test whether the average total cholesterol level for U.S. adults has decreased from the 2013-2014 level of 189 mg/dL. Define the parameter in context.
b. One of the conditions needed for us to perform inference for the population mean total cholesterol level for U.S. adults is that we have a random sample from the population. Check the other condition for inference.
c. Assume that the conditions for inference are satisfied. Calculate the test statistic. Be sure to show your work.
d. Which of the following should be used to calculate the p-value?
   - `pnorm(test statistic value, mean = 0, sd = 1, lower.tail = FALSE)`
   - `pnorm(test statistic value, mean = 0, sd = 1, lower.tail = TRUE)`
   - `pt(test statistic value, df = 79, lower.tail = FALSE)`
   - `pt(test statistic value, df = 79, lower.tail = TRUE)`
e. The p-value was calculated to be greater than 0.10. What would you tell the researcher about their findings?

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21. **Squirrel!** In many parts of the northern United States, two color variants of the Eastern Gray Squirrel—gray and black—are found in the same habitats. A scientist studying squirrels in a large forest wonders about the difference in the sizes of the two color variants. The scientist collects random samples of 40 squirrels of each color from a large forest and weighs them. The table below gives summary statistics for the two groups of squirrels.

	Mean	SD
Black squirrels	20.3 ounces	2.1 ounces
Gray squirrels	19.2 ounces	1.9 ounces

a. The scientist's friend says, "We should treat these as dependent samples since there are 40 black squirrels and 40 gray squirrels." Briefly explain why the friend is not correct (and that we should treat these as independent samples).
b. The independence conditions are satisfied for inference. We don't have histograms of the weights of the black squirrels and the weights of the gray squirrels, so we don't know if the weights come from populations that have normal distributions. Why can we continue with inference?
c. Calculate a 95% confidence interval for the difference in the mean weights of black and gray Eastern Gray Squirrels in this forest. For your reference, the critical value is 1.991.
d. Suppose that prior to conducting research, the scientist thought that black squirrels weigh more than gray squirrels. Does your confidence interval from part (c) support what the scientist thought? Clearly state "yes" or "no" and briefly explain your reasoning.
